An Ensemble and Deep Learning Based Approach for SMS Spam Classification

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***Abstract*— The Short Message Service (SMS) remains a critical medium of communication despite the arrival of newer messaging platforms; nevertheless, the pervasive dissemination of spam messages remains an endemic problem for users across the world. This paper introduces a discerning method to detect SMS spam with ensemble-based models coupled with deep embedding and transformer models to address the subtlety of natural language processing and linguistic ambiguity. We performed a thorough analysis on a data set that was typified by imbalance with 5,575 messages obtained from Kaggle, using exhaustive preprocessing techniques involving cleaning of text, stop word removal, and lemmatization to improve quality of input and reduce noise. Our experimental results demonstrate outstanding performance for different configurations of the model: the Random Forest coupled with TF-IDF achieved 98% accuracy with a 99% F1 score, highlighting the continued relevance of conventional machine learning methods when accordingly optimized. The ALBERT transformer model achieved 99.73% accuracy after 15 epochs of training, testifying to the efficacy of contextual embeddings in this classification task. The better performance demonstrated by conventional and novel models supports our hybrid approach to spam detection. The research conclusively proves that blend of machine learning techniques with innovative deep learning architectures and advanced preprocessing techniques yields an efficient system to address the endemic problem of SMS spam, potentially improving user experience and security on billions of daily text messages worldwide.**

Keywords—SMS Spam Classification; Natural Language Processing; Ensemble Learning; TF-IDF; Sentence Transformers; Mobile Security

I. Introduction

Spam refers to unsolicited electronic messages sent by spammers with the intent of causing psychological[1] or financial harm to victims. It appears in a variety of forms, including web spam, review spam, short message service (SMS) [2]spam, and email spam[3]. In survey spam, spammers manipulate reviews by providing false positive (FP) feedback[4]. SMS spam sent via text message not only annoys the senders but can also cost the service provider money. Email spam[5] is generally junking advertisements or content sent by individuals who do not have a valid relationship with the recipient. SMS spam is distributed in a variety of ways, including infected devices, hacked servers, phishing schemes, and unauthorized collection of users' phone numbers[6].

In 2020, Vodafone Italy faced a major SMS phishing attack where users were tricked into sharing sensitive information, highlighting the severe impact of SMS spam on individuals and organizations. Although prior studies using datasets like Kaggle and UCI with classifiers such as Random Forest and Random Tree achieved moderate success, they were limited by small datasets, low accuracy, and ineffective feature extraction. To address these gaps, this research applies NLP, ensemble-based methods, and transformer-based models to improve SMS spam detection accuracy.

To secure users' information and provide a better experience for all users. The limitations discussed in this paper have been addressed through the implementation of a new approach that combines ensemble-based methods-random forest and AdaBoost-and transformer-based models-Span BERT and ALBERT for the task of SMS spam classification[7]. We use advanced encoding techniques such as Sentence Transformers and TF-IDF to improve semantic understanding and feature extraction. With a focus on these methodologies, we try to develop a model that will be efficient in distinguishing spam messages from the valid ones[8], even in cases where there are diverse datasets and limited resources. Besides, we have conducted extensive validation on a Kaggle dataset with 5,575 SMS messages to further validate the efficiency and reliability of the proposed models.

The key contributions of this study can be summarized as follows:

- To apply diverse NLP techniques to preprocess the unstructured textual dataset.

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- To utilize feature engineering techniques such as TF-IDF and sentence transformers to convert the textual data into fixed vector representation,
- To employ transformers based models such as SpanBERT and ALBERT for SMS spam classification.
- Carried out extensive experiments to assess the performance of the proposed models.

Section 2 reviews related work, Section 3 details the methodology, Section 4 presents results, and Section 5 concludes with findings and future work.

II. Related Work

In a study[9], researchers proposed text classification methods for SMS spam detection using machine learning algorithms on a Kaggle dataset of 611 messages. Experiments with the Weka tool showed Random Forest outperformed other models, but limitations included the small dataset size, high implementation cost, and oversimplified classification methods. Alternative strategies, which include a double pass filter, that integrates KNN[10] with rough set principles, SVM and Naïve Bayes showed promising results, while recent work with Random Forest and Naïve Bayes on the H2O platform highlighted the need for more efficient systems. Research [11], in Pakistan, studies on Urdu spam and scam messages used SVM, Naïve Bayes, and Decision Trees (J48) on a manually collected dataset of 1050 messages. However, concerns were raised about dataset bias and limited coverage of fraudulent message types. Nevertheless, research produced encouraging results, which suggested a significant impact on reducing user risk for SMS fraud in novel Urdu [12]. With rising SMS use, especially in Romanized Urdu, one study applied an SVM-based approach on a dataset of 8107 messages, achieving 82.6% accuracy. However, dependence on a single user-related model introduced bias and limited exploration of better alternatives. A mobile-based ML approach [13] used four features (size, monograms, diagrams, class) on 4600 SMS messages, testing NB, MLP, and C4.5. Naïve Bayes achieved ~80% accuracy with low computational cost, though results relied heavily on dataset quality. Another study[14], developed a spam filtration technique using the term frequency-inverse document frequency (TF-IDF) and Random Forest (RF) algorithm for functional extraction. The dataset included 4989 English SMS messages from UCI Machine Learning Repository[15]. The study deal with data imbalance using further screen measurements and found that RF with TF -IDF gained an accuracy of 0.98 and F1-measurement of 0.97. The study suggested that incorporating additional functions, such as the length of the message, may further increase performance. A comprehensive analysis [16] the paper examines detection of SMS spams in different geographical areas, and evaluated several machines learning techniques, including Naive Bayes, SVM, decision tree, logistic region and random forests. Most of the approach depend on the message material for the extraction of functions. The study identified weaknesses in current spam detection methods, such as datasets, such as UCI SMS Spam collection to assess performance[17]. Challenges included local languages, summary and handling of incomplete data, all of which affect classification accuracy[18]. The review emphasized the requirement for diverse data sets and standardized evaluation matrix to increase the strength of SMS Spam detection systems.[19]

The study [20] analyzed 3560 English SMS messages (UCI dataset) using TF-IDF features and Random Forest, achieving 98% accuracy and 0.97 F1-Score. Despite dataset imbalance favoring ham messages, results showed strong performance, with recommendations for further improvement.

Current telecom spam filters (e.g., Verizon, AT&T, T-Mobile) and built-in phone tools (Apple, Google) mainly use rule-based or basic ML models with keyword matching and sender scoring. These systems face high false positives, poor adaptability to evolving spam, weak multilingual handling, scalability issues, and privacy concerns. Unlike them, our proposed method leverages ensemble learning, transformer-based contextual understanding, and advanced feature engineering for higher accuracy and adaptability.

III. Proposed Research Methodology

The process includes text pre-processing (cleaning, lemmatization, stop word removal) and feature extraction using TF-IDF and Sentence Transformers. Deep learning models (SpanBERT, ALBERT) and ensemble (Random Forest, AdaBoost) are trained, and performance is measured using accuracy, precision, recall, and F1-score.

A. Performance Evaluation Measures

The experimental evaluation employed comprehensive performance metrics specifically selected to address the challenges inherent in imbalanced classification tasks, where traditional accuracy measurements prove insufficient for meaningful assessment. Precision quantified the proportion of correctly identified spam messages among all messages classified as spam, serving as a critical metric for minimizing false positive rates that could result in legitimate message blocking and user dissatisfaction. Recall measured the proportion of actual spam messages correctly identified by the classification system, ensuring comprehensive spam detection coverage and minimizing false negative rates that would allow malicious content to reach users. The F1-score provided a harmonic mean between precision and recall, offering a balanced evaluation metric particularly valuable for imbalanced datasets where optimizing accuracy alone could result in biased models favoring the majority class, while confusion matrices enabled detailed analysis of classification performance patterns and error distribution across both spam and ham categories.

B. Experimental Setup

The computational experiments were executed on a high-performance system equipped with Intel Core i7-12700K processor operating at 3.6GHz base frequency, 32GB DDR4 RAM, and NVIDIA RTX 3080 graphics processing unit with 10GB VRAM, providing adequate computational resources for both ensembles learning algorithms and transformer-based model training. The class imbalance problem was systematically addressed through stratified k-

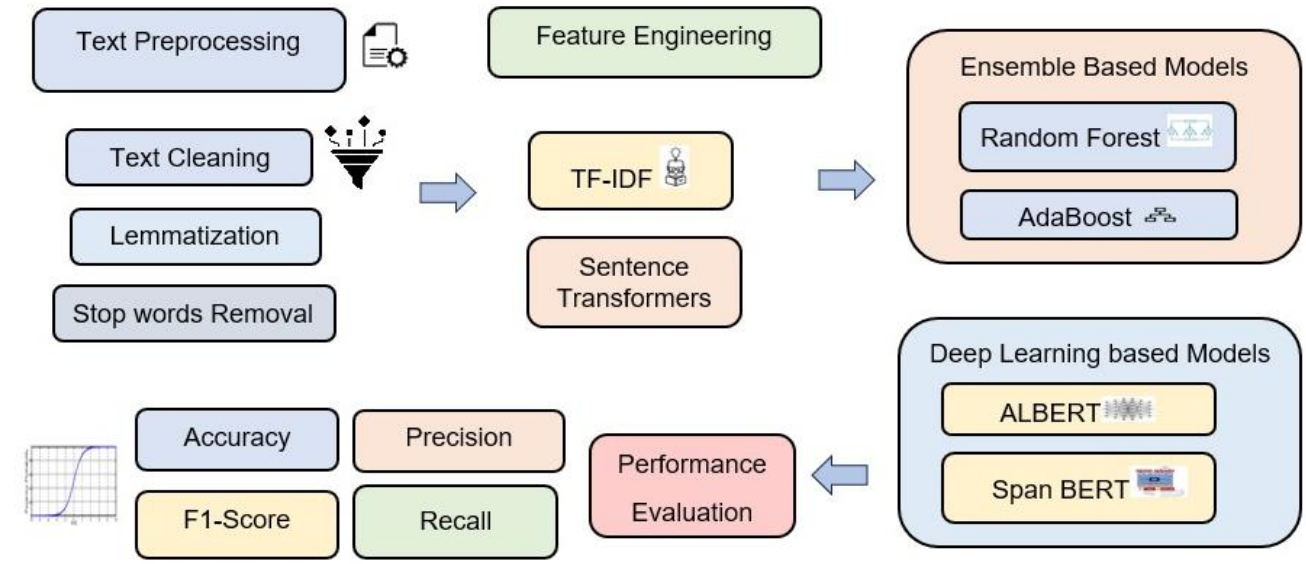

Fig. 1: Overview of the proposed framework for detection of SMS spam using ensemble and deep learning models

fold cross-validation maintaining consistent class distribution ratios across training and validation sets, while ensemble methods inherently managed imbalance through Random Forest's balanced sampling strategies and AdaBoost's adaptive focus on misclassified instances, with transformer models employing class weighting mechanisms during fine-tuning phases. Deployment considerations encompass significant computational constraints for mobile device implementation where transformer models require substantial memory and processing resources that may be prohibitive for real-time analysis, coupled with evolving spam tactics employing sophisticated adversarial text generation, semantic obfuscation techniques, and dynamic content modification strategies that necessitate continuous model adaptation and retraining to maintain classification effectiveness against emerging threat patterns.

C. Dataset Description

The experimental validation was conducted using a publicly available SMS spam dataset obtained from Kaggle, comprising 5,575 messages with a pronounced class imbalance characteristic of real-world spam detection scenarios. The dataset contained 4,827 legitimate messages (ham) representing 86.6% of the corpus and 748 spam messages constituting 13.4% of the total collection. Spam messages exhibited diverse patterns including promotional advertisements with phrases like "URGENT! You have won a £1000 cash prize," phishing attempts requesting sensitive information such as "Update your bank details immediately," and fraudulent schemes promising unrealistic monetary rewards. Legitimate messages encompassed everyday conversational content, appointment confirmations, family communications, and business-related exchanges, demonstrating natural linguistic patterns with contextual coherence and personal communication characteristics that distinguished them from the commercially oriented and urgency-driven language patterns prevalent in spam content

D. Text Preprocessing

Text preprocessing is vital for the refinement and analysis of text data, converting raw unstructured text into actionable insights. NLP, sentiment analysis are examples of text analysis techniques used in comparison of the patterns, relationships, and themes. Tokenization, stemming, and lemmatization are data cleaning techniques that, clean and standardize data. This process increases the quality and accuracy of research outcomes across disciplines.

1) Text Cleaning

Text cleaning is essential for improving quality and uniformity of the data. Data cleaning mainly consists of removing the non-textual component and converting text to lowercase for uniformity. This helps in normalizing and correcting spelling errors, as well as the reduction of redundancy. Ultimately, the management of punctuation provides data that is clean and consistent for a specific type of analysis. This process makes the text data efficient

2) Stop Words Removal

Stop words are worth little meaning, such as 'and' or 'the', and are removed during preprocessing for noise reduction and further purification into meaningful terms. This reduces the dimensionality and size of data, improving the text analysis and performance in Machine Learning. Removing stop words becomes noise free and streamlines the whole analysis process. It makes the process fast and accurate.

3) Lemmatization

Lemmatization reduces words to their basic size (lemma) form using context and grammar, unlike stemming. It reduces the complexities in the text, groups similar words together, and improves consistency for better pattern recognition and analysis. It is important for accurate NLP models and effective text analysis.

E. Feature Engineering

Feature engineering involves transforming textual data into fixed numeric representation. We applied two NLP approaches for encoding the SMS textual data, including TF-IDF and Sentence Transformers.

1) TF-IDF (TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF is a measure in NLP that evaluates a phrase's significance in a document relative to a corpus. It combines term frequency (TF), indicating a word's relevance within a document, and inverse document frequency (IDF), reflecting its rarity throughout the corpus. High TF-IDF ranks or scores highlight unique and significant terms. Widely used in tasks like textual content summarization and key-word extraction, TF-IDF enhances document analysis and retrieval by prioritizing key terms.

$$TF(t,d) = \frac{f(t,d)}{total\ \ (d)} \quad (1)$$

Where, TF is a time period t in a document d. It measures how often a time period appears inside the documents, normalized with the total number of words, highlighting the term's relative significance within that document. This normalization prevents longer documents from skewing the evaluation because of word count variations and differences. In spam-ham SMS analysis, TF helps in identifying frequent occurrence of words in messages allowing significant insights when combined with IDF to distinguish among common and specific terms across the dataset.

$$IDF(t) = log(\frac{N}{doc_{freq}(t)}) \quad (2)$$

$$TF - IDF(t) = TF(t) + IDF(t) \quad (3)$$

This equation is used in information retrieval and text mining to calculate the Inverse Document Frequency (IDF) of a term t. Here, N represents the total number of documents in the corpus, while doc_freq(t) denotes the number of documents that contain the term t. The IDF value helps assess the importance of a term within a specific document relative to the entire collection, with higher values indicating that the term is rare across documents, thus potentially more significant for distinguishing relevant content. Subsequently, by multiplying TF and IDF, we get a final TF-IDF score for each term within a document.

2) Sentence Transformer

A Sentence Transformer is an NLP model that encodes sentences into dense numerical vectors, capturing their semantic meaning. Unlike word embedding's, it focuses on whole sentences using transformer architectures like BERT, variants of BERT i.e. ALBERT, span BERT. These models are key for semantic search, sentence similarity, and text classification, where NLP accuracies can be improved based on richer contextual representations.

$$CosineSimilarity(c1, c2) = \frac{c1 \cdot c2}{\|c1\|\|c2\|} \quad (4)$$

The equation (4) finds the cosine angle between two quantities: c1 and c2, and c1 and c2 in terms of sentence embeddings. In the numerator is the dot product of the two, meaning their directional alignment, while the denominator normalizes the result against the magnitudes of the two respectively. A cosine similarity value closer to 1 indicates that the sentences are similar in meaning, while a value closer to 0 suggests they are dissimilar.

F. Proposed Models for SMS Spam Detection

In this study, we applied diverse machine learning and deep learning models to detect SMS spams for noise textual data. The details of proposed models are as follows:

1) Ensemble-bassed Machine Learning Models

Machine learning models analyze data, identify patterns, and make predictions from labeled datasets. In this study, we use two ensemble models—Random Forest and AdaBoost—to classify SMS messages as spam or ham.

a) Random Forest

Random Forest is an ensemble method that builds multiple trees on random subsets of data and features, reducing overfitting. It is effective for classification, handles high-dimensional data well, and performs reliably with minimal hyperparameter tuning.

$$\hat{y} = argmax_c \sum(T_i(x) == c) \quad (5)$$

The equation shows the prediction mechanism of a random forest model. Here, (y) is the predicted class obtained by maximizing the sum of votes across all trees (T_i), where (i(x) == c) indicates that the (i)-th tree predicts class (c). The model aggregates these votes to determine the final class (y) for the input (x).

b) AdaBoost

AdaBoost (Adaptive Boosting) is an ensemble method that combines weak classifiers, often decision trees, to form a strong model. It assigns higher weights to misclassified instances, forcing later classifiers to focus on difficult cases. Final predictions are made through weighted voting, giving more influence to accurate classifiers. The final model:

$$F_T(x) = \sum_{t=1}^{T} f_t(x) \quad (6)$$

aggregates weak classifiers, each assigned a coefficient a_t to minimize training error:

$$E_t = \sum_i E[F_{t-1}(x_i) + a_t h(x_i)] \quad (7)$$

Here the misclassified instances are focused, which enhances the accuracy and robustness in classification tasks.

2) Deep Learning Models

Deep Learning is a subset of Machine Learning models inspired by the human brain and based on artificial neural networks. It confined to different layers of neurons to get higher levels of features form data. The study involves two diverse variations of BERT models. The details of each model are as follows:

a) ALBERT

ALBERT (A Lite BERT) is a lightweight variant of BERT that reduces memory and training costs while maintaining performance. Its transformer-based architecture effectively

extracts semantic and contextual relationships in short SMS texts, making it suitable for spam vs. ham classification.

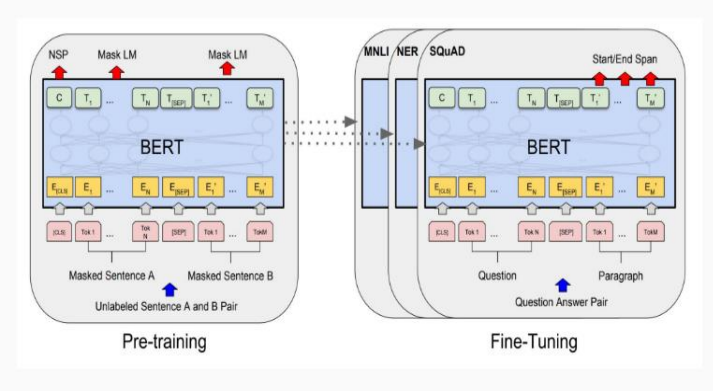

Fig. 2: The Architecture of ALBERT: Pre-training and Fine-Tuning

The following equation (8) shows the probability P_i of a special class i (spam or ham) according to the ALBERT model in the case of SMS classification.

$$P_i = \frac{k_i}{\sum_j k_j} \tag{8}$$

Where the term k_i denotes the score or output for the class i while $\sum_f k_f$ is the sum of scores across all possible classes.

b) Span BERT

SpanBERT is a BERT variant that predicts text spans, improving contextual understanding. For SMS spam detection, it captures semantic patterns effectively, making it useful for identifying phishing and scam messages.

$$L(x_i) = L_{MLM}(x_i) + L_{SBO}(x_i) = -logP(x_i|X_i) - logP(x_i|Y_i) \tag{9}$$

Where, $L(x_i)$ is the overall loss for the input x_i, combining two loss components and $L_{MLM}(x_i)$ denotes the loss from the masked language model (MLM), capturing how well the model predicts the masked parts of the input x_i given the contextX_i. Moreover, the term $L_{SBO}(x_i)$ is the loss from the span-based objective (SBO), which assesses the model's ability to predict x_i based on additional contextual information Y_i and $P(x_i|X_i)$ and $P(x_i|Y_i)$ represents the probability of the input x_i given its context X_i in the MLM framework and input x_i given an alternative context Y_i in the SBO framework respectively.

IV. Results and Discussion

The proposed framework results show the impact of encoding methods on SMS spam classification. With Sentence Transformer encoding, AdaBoost outperformed Random Forest, achieving 97% accuracy, 98% precision, 98% recall, and a 98% F1-score, while Random Forest reached 96% accuracy, 96% precision, and a 98% F1-score. Using TF-IDF, Random Forest achieved the best performance with 98% accuracy, 98% precision, and a 99% F1-score, whereas AdaBoost performed slightly lower with 95% accuracy and 97% F1-score. These findings highlight the critical role of selecting appropriate encoding methods to maximize performance.

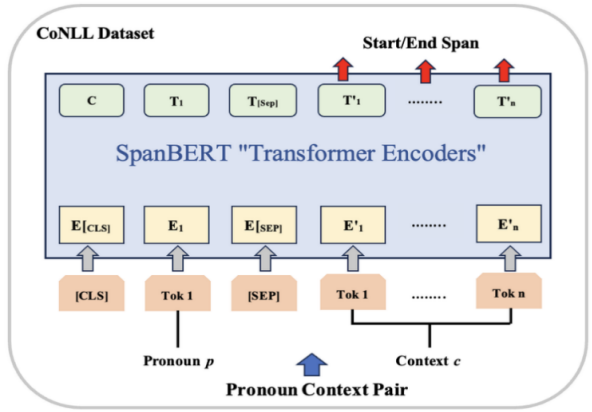

Fig. 3: The architecture of SpanBERT for Contextual Understanding

TF-IDF encoding keeps at its best with the Random Forest model, giving the highest F1-score of 99%. On the other hand, Sentence Transformer encoding is suitable for AdaBoost better, providing strong performance across all metrics. This analysis indicates how different encoding methods can perform well with ensemble models for SMS spam classification, contributing flexibility and strong performance.

Table 1: Results of Ensemble Based Models with Sentence Transformer and TF-IDF over SMS Spam Classification Dataset

Model	Encoding Methods	Acc %	Precision (%)	Recall (%)	F1 Score (%)
AdaBoost	Sentence Transform	93.22	90.31	89.79	90.25
	TFIDF	95.84	95.54	66.58	78.03
Random Forest	Sentence Transform	97.56	96.60	77.54	87.74
	TFIDF	98.11	98.12	86.52	92.95

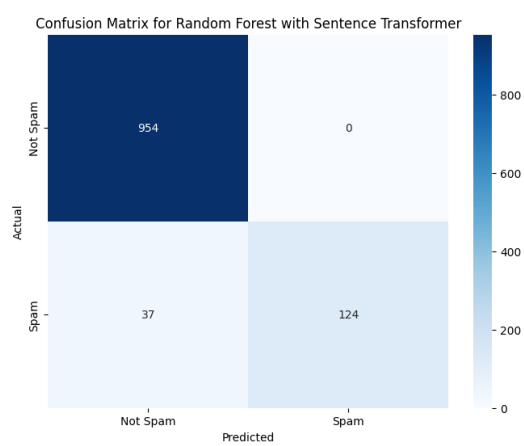

Fig. 4 Confusion Matrix for Random Forest

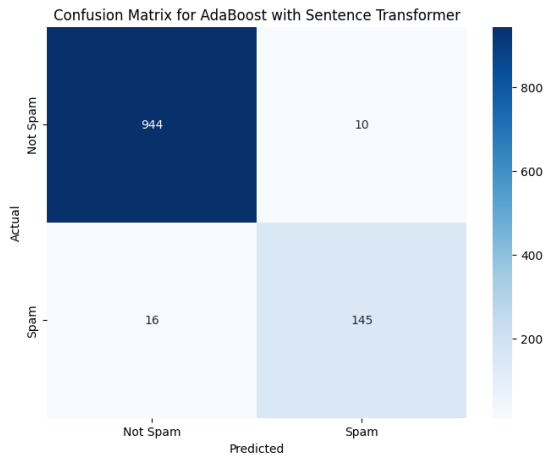

Fig. 5: Confusion Matrix for AdaBoost

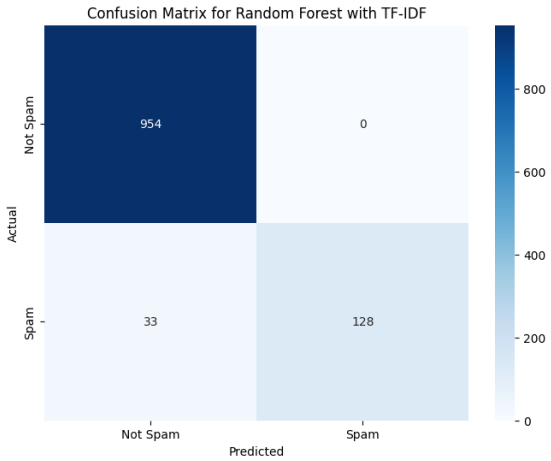

Fig 6: Confusion Matrix for Random Forest

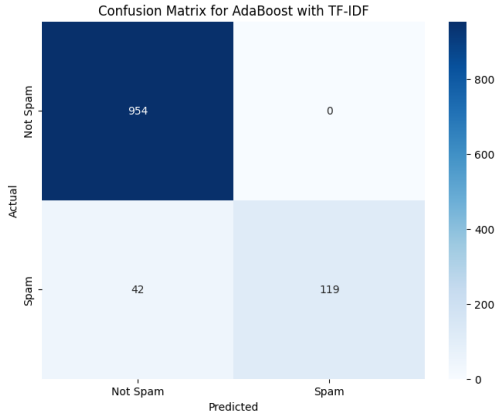

Fig. 7: Confusion Matrix for AdaBoost

The confusion matrices (Figure 4, 5, 6, 7, 9 and 10) illustrate the classification performance of the proposed models. As observed, the Random Forest with TF-IDF emphasizes precision by reducing false positives, while the transformer-based model (e.g., ALBERT) provides a better balance between precision and recall. This trade-off highlights the practical implications of model selection in real-world spam filtering applications.

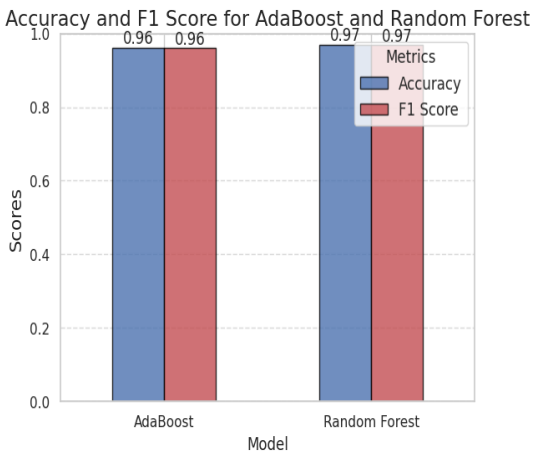

Fig. 8: Bar Chart of Accuracy and F1-Score for AdaBoost and Random Forest

Table 2: Results of Albert over SMS Spam Classification

Model	Epochs#	Accuracy %	Precision %	Recall %	F1-Score %
Albert	1	99.28	99.27	99.28	99.28
	5	99.28	99.28	99.38	99.27
	10	99.55	99.55	99.55	99.55
	15	99.73	99.73	99.73	99.73
	20	99.73	99.73	99.73	99.73

According to Table 2, with the TF-IDF encoding-based transformer model achieves 99.28% peak accuracy at 1 epoch. In contrast, the ALBERT model outperforms with a 99.55% accuracy at 10 epochs along with 99.55% precision, recall as well as F1-Score. Continuing training to 15 epochs, ALBERT further improves to an accuracy of 99.73%, maintaining the same precision, recall, and F1-Score.

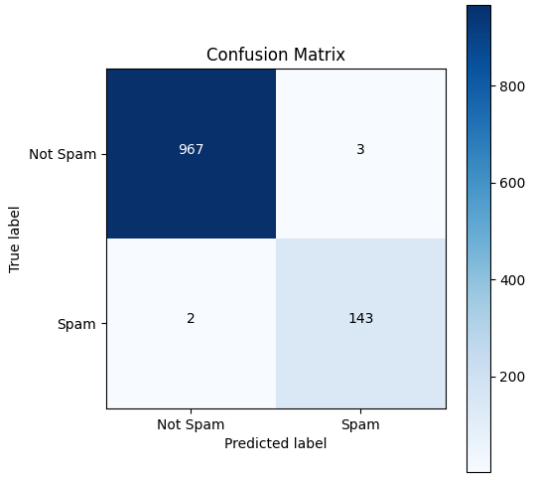

Fig. 9: Confusion Matrix for ALBERT

According to Table 3, the transformer model achieves the highest accuracy of 98.47% at 1 epoch, with precision, recall and F1 score values closely aligned. The model continues to improve through training up to 5 epochs, when its accuracy rises to 99.46% along with precision and recall values.

Table 3: Results of Span BERT over SMS Spam Classification

Model	Epochs#	Accuracy %	Precision %	Recall %	F1-Score %
Span BERT	1	98.47%	98.50%	98.47%	98.43%
	5	99.46%	99.46%	99.46%	99.45%
	10	98.11%	98.29%	98.11%	98.16%
	15	99.19%	99.20%	99.12%	99.18%
	20	99.46%	99.46%	99.46%	99.45%

On the other hand, the Span BERT model, shows an accuracy of 98.11% at 10 epochs, but excels at 15 epochs, gaining an accuracy of 99.19% and sustains a robust result of 99.46% accuracy at the 20th epoch, showing stable and reliable results across the epochs. Overall, the results highlight the effectiveness of transformer-based models in achieving high performance in classification tasks.

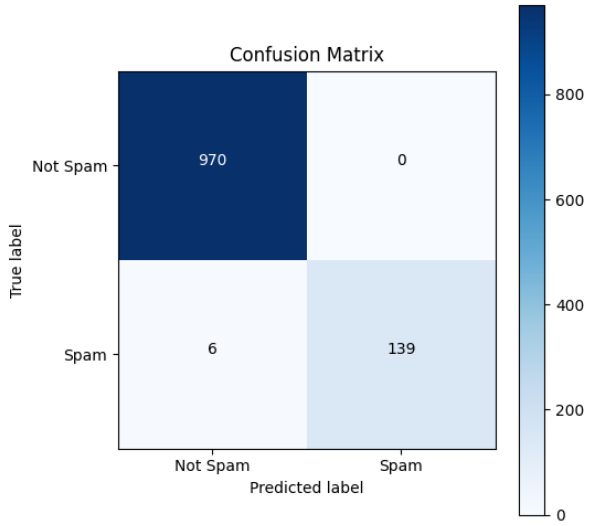

Fig. 10: Confusion Matrix for Span BERT

V. Conclusion & Future work

This study examined ensemble models (Random Forest, AdaBoost) and deep learning methods (ALBERT, SpanBERT) for SMS spam classification. Using advanced preprocessing and feature engineering (TF-IDF, Sentence Transformer embeddings), Random Forest with TF-IDF achieved 98% accuracy and 99% F1-score, while ALBERT excelled with 99.73% accuracy at 15 epochs. These results show that combining diverse encodings with robust models enhances spam detection accuracy and reliability. Future work may expand datasets with multilingual and region-specific SMS to improve generalizability and develop lightweight, real-time detection systems for resource-constrained devices. Exploring advanced feature extraction methods such as sentiment, syntax, and user behavior can improve accuracy. Hybrid models combining ensembles with NAS or transfer learning may further enhance performance. Incorporating adversarial training strengthens robustness against attacks, while XAI improves transparency. Long-term evaluation against evolving spam patterns ensures secure and effective SMS spam classification.

The confusion matrix analysis highlights key trade-offs between precision and recall in spam detection. Random Forest with TF-IDF achieved high precision (98%) but lower recall, underscoring the challenge of minimizing false positives while ensuring broad spam coverage. Such trade-offs are critical in deployments where blocking legitimate messages impacts user trust. In contrast, transformer models, particularly ALBERT, showed more balanced performance with symmetric accuracy and fewer misclassifications across spam and ham. Overall, ensemble methods suit applications prioritizing precision and low false positives, while transformers are better for scenarios requiring balanced precision and recall.

References

[1] J. A. Orr and R. J. King, "Mobile phone SMS messages can enhance healthy behaviour: a meta-analysis of randomised controlled trials," Taylor & Francis, vol. 9, no. 4, pp. 397–416, Nov. 2015, doi: 10.1080/17437199.2015.1022847.

[2] J. Brown, B. Shipman, R. V.- Computer, and undefined 2007, "SMS: The short message service," ieeexplore.ieee.org, Accessed: Apr. 28, 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/4404822/

[3] R. L. J.-J. of I. Science and undefined 1989, "Minimizing information overload: the ranking of electronic messages," journals.sagepub.com, vol. 15, no. 3, pp. 179–189, 1989, doi: 10.1177/016555158901500306.

[4] R. Shaw, H. B.-H. informatics journal, and undefined 2012, "Short message service (SMS) text messaging as an intervention medium for weight loss: a literature review," journals.sagepub.com, vol. 18, no. 4, pp. 235–250, Dec. 2012, doi: 10.1177/1460458212442422.

[5] G. C.-F. and T. in Information and undefined 2008, "Email spam filtering: A systematic review," nowpublishers.com, doi: 10.1561/1500000006.

[6] A. Bhowmick, S. H.-A. in Electronics, C. and, and undefined 2017, "E-mail spam filtering: a review of techniques and trends," Springer, vol. 443, pp. 583–590, 2018, doi: 10.1007/978-981-10-4765-7_61.

[7] T. Guzella, W. C.-E. S. with Applications, and undefined 2009, "A review of machine learning approaches to spam filtering," Elsevier, Accessed: Apr. 28, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S095741740900181X

[8] S. Delany, M. Buckley, D. G.-E. S. with Applications, and undefined 2012, "SMS spam filtering: Methods and data," Elsevier, Accessed: Apr. 28, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417412002977

[9] T. A.-P. of the C. name and undefined 2020, "A study of sms spam using machine learning," researchgate.netT AsgherProceedings of the Conference name, 2020•researchgate.net, Accessed: Apr. 26, 2025. [Online]. Available: https://www.researchgate.net/profile/Tayba-Asgher/publication/345311834_A_Study_of_SMS_Spam_using_Machine_Learning/links/5fa2d226a6fdcc0624125680/A-Study-of-SMS-Spam-using-Machine-Learning.pdf

[10] M. Ahmed, H. Khan, T. Iqbal, F. K. Alarfaj, A. Alomair, and N. Almusallam, "On solving textual ambiguities and semantic vagueness in MRC based question answering using generative pre-trained transformers," PeerJ Comput Sci, vol. 9, p. e1422, Jul. 2023, doi: 10.7717/PEERJ-CS.1422/SUPP-13.

[11] D. Suleiman, G. Al-Naymat, M. I.-I. Journal, and undefined 2020, "Deep SMS spam detection using H2O platform," researchgate.netD Suleiman, G Al-Naymat, M ItriqInternational Journal, 2020•researchgate.net, Accessed: Apr. 26, 2025. [Online]. Available: https://www.researchgate.net/profile/Dima-Suleiman/publication/345312255_Deep_SMS_Spam_Detection_using_H2O_Platform/links/5fa2e01892851cc286939a68/Deep-SMS-Spam-Detection-using-H2O-Platform.pdf

[12] "Detection of Roman Urdu fraud/spam SMS in Pakistan... - Google Scholar." Accessed: Apr. 26, 2025. [Online]. Available: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=Detection+of+Roman+Urdu+fraud%2Fspam+SMS+in+Pakistan+Using+Machine+Learning+authored+by+Muhammad+Ayaz1%2C+Sarwat+Nizamani2%2C+Aftab+Ahmed+Chandio3+and+Kirshan+Kumar+Luhana4&btnG=

[13] M. Khan, S. Ayub, M. Khan, … M. A.-P. J. of, and undefined 2019, "Machine learning for content-based detection of spam sms in local languages: a preliminary classification of romanized urdu messages," search.proquest.comMS Khan, SI Ayub, MA Khan, M Afaq, F TilaPakistan Journal of Science, 2019•search.proquest.com, Accessed: Apr. 26, 2025. [Online]. Available: https://search.proquest.com/openview/10d8759c0ecb0587af12866785d3a0af/1?pq-origsite=gscholar&cbl=1616340

[14] "SMS Spam Detection Using Simple Message Content Features... - Google Scholar." Accessed: Apr. 26, 2025. [Online]. Available: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=SMS+Spam+Detection+Using+Simple+Message+Content+Features+Dr.+Ghulam+Mujtaba1%2C+Majid+Yasin2&btnG=

[15] J. Nawal, G. Arshad, M. Ahmed, M. Shahid, and H. Khan, "Medical Intent Classification Using Ensemble and Deep Learning Models," researchgate.netJ Nawal, G Arshad, M Ahmed, MMA Shahid, HU Khanresearchgate.net, Accessed: Apr. 28, 2025. [Online]. Available: https://www.researchgate.net/profile/Ijist-Jr/publication/389322102_Medical_Intent_Classification_Using_Ensemble_and_Deep_Learning_Models/links/67be873c96e7fb48b9cde40e/Medical-Intent-Classification-Using-Ensemble-and-Deep-Learning-Models.pdf

[16] K. Hanif, H. G.-Irjmets. com, and undefined 2021, "Detection of Sms Spam and Filtering By Using Data Mining Methods: Literature Review," academia.eduK Hanif, H GhousIrjmets. com, 2021•academia.edu, Accessed: Apr. 26, 2025. [Online]. Available: https://www.academia.edu/download/97289305/1628083238.pdf

[17] M. Ahmed, H. U. Khan, and E. U. Munir, "Conversational AI: An Explication of Few-Shot Learning Problem in Transformers-Based Chatbot Systems," IEEE Trans Comput Soc Syst, vol. 11, no. 2, pp. 1888–1906, Apr. 2024, doi: 10.1109/TCSS.2023.3281492.

[18] E. Dada, J. Bassi, H. Chiroma, S. A.- Heliyon, and undefined 2019, "Machine learning for email spam filtering: review, approaches and open research problems," cell.com, Accessed: Apr. 28, 2025. [Online]. Available: https://www.cell.com/heliyon/fulltext/S2405-8440(18)35340-4

[19] S. N.-J. of E. T. and Innovative and undefined 2018, "Mobile SMS Spam Detection using Machine Learning Techniques," researchgate.netS NagreJournal of Emerging Technologies and Innovative Research, 2018•researchgate.net, Accessed: Apr. 26, 2025. [Online]. Available: https://www.researchgate.net/profile/Samadhan-Nagare/publication/371292744_Mobile_SMS_Spam_Detection_using_Machine_Learning_Techniques/links/647dadfbb3dfd73b7767dab9/Mobile-SMS-Spam-Detection-using-Machine-Learning-Techniques.pdf

[20] N. Sjarif, N. Azmi, S. Chuprat, … H. S.-P. C., and undefined 2019, "SMS spam message detection using term frequency-inverse document frequency and random forest algorithm," ElsevierNNA Sjarif, NFM Azmi, S Chuprat, HM Sarkan, Y Yahya, SM SamProcedia Computer Science, 2019•Elsevier, Accessed: Apr. 26, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050919318617